Math 110, Fall 2026 **Practice Midterm 2** **Prof. Hammond**

Preparation for Friday, October 30, 2026

NAME: _____ **SID:** _____

LAST NAME, OTHER NAMES (IN BLOCK CAPITALS)

This practice paper has four questions worth 40 points in total. It may take longer than 50 minutes; the purpose is to practise mathematical reasoning at approximately the level of the midterm, not to predict its questions.

NOTATION

For a linear map T, its null space and range are denoted by $\operatorname{null} T$ and $\operatorname{range} T$. All bases are ordered when coordinates or matrices are requested.

INSTRUCTIONS

1. We will be scanning your exam and using Gradescope. Please write only within the black borders of each page, and write your name in the space provided on each question page.
2. Please write a concise answer to each part in the space provided, including calculations or reasoning as well as the final answer.
3. You may quote without proof any theorem or example from the text, lecture notes, or worksheets, provided the result is clearly stated, unless the problem asks you to prove it.
4. Extra space for each question is provided on the back of the preceding page. Use that page only for work on the indicated question.
5. The back of the final page may be used for rough work on any question. It will not be graded.
6. Attempt the questions before consulting the solutions.

Extra space for QUESTION 1 ONLY

Name: _____

1. A quotient of a polynomial space. Let $V = \mathcal{P}_3(\mathbb{R})$ and define $E : V \to \mathbb{R}^2$ by $E(p) = (p(0), p(1))$. Put $U = \operatorname{null} E$.

a) (2) Prove that E is linear and surjective.

b) (3) Find a basis for U.

c) (3) Prove that the formula $\widetilde{E}(p + U) = E(p)$ defines a linear isomorphism from V/U to $\mathbb{R}^2$.

d) (2) Give a basis for V/U and specify the images of its elements under $\widetilde{E}$.

Extra space for QUESTION 2 ONLY

Name: _____

2. Dual maps. Let V, W, Z be finite-dimensional real vector spaces, and write $T : V \to W$ and $S : W \to Z$ for linear maps. Denote their dual maps by $T^* : W^* \to V^*$ and $S^* : Z^* \to W^*$.

a) (2) Give the definition of T^* and prove that it is linear.

b) (3) Prove that $(S \circ T)^* = T^* \circ S^*$.

c) (3) Prove that if T is surjective, then T^* is injective.

d) (2) Suppose $V = \mathbb{R}^2$, $W = \mathbb{R}$, and $T(x, y) = 2x - y$. If $\ell \in W^*$ is defined by $\ell(t) = 3t$, compute $T^*\ell$.

Extra space for QUESTION 3 ONLY

Name: _____

3. Upper-triangularization. Let V be a nonzero finite-dimensional complex vector space, and let $T : V \to V$ be linear. You may use the theorem proved in lectures that every operator on a nonzero finite-dimensional complex vector space has an eigenvalue. Prove, by induction on $\dim V$, that some basis of V represents T by an upper-triangular matrix.

a) (2) Establish the assertion when $\dim V = 1$.

b) (3) For $\dim V > 1$, choose an eigenvalue λ of T and put $U = \text{range}(T - \lambda I)$. Prove that U is invariant under T and that $\dim U < \dim V$.

c) (2) Explain how the induction hypothesis yields a basis $u_1, \ldots, u_m$ of U satisfying $Tu_j \in \text{span}(u_1, \ldots, u_j)$ for each j.

d) (3) Extend this to a basis $u_1, \ldots, u_m, v_1, \ldots, v_r$ of V and finish the proof using $Tv_j = (T - \lambda I)v_j + \lambda v_j$.

Extra space for QUESTION 4 ONLY

Name: _____ **4. A consequence of the**

minimal polynomial. Let V be a nonzero finite-dimensional vector space over $\mathbb{R}$, let $T : V \to V$ be linear, and let m_T denote its minimal polynomial.

a) (3) Suppose $m_T(0) \neq 0$. Prove that T is invertible by expressing its inverse as a polynomial in T.

b) (3) Suppose instead that $m_T(0) = 0$. Prove that T is not invertible. You may use the defining minimality of m_T.

c) (2) Let $R : \mathbb{R}^2 \to \mathbb{R}^2$ be given by $R(x, y) = (-y, x)$. Find m_R.

d) (2) Use (a) to express R^{-1} in terms of R, and check your answer directly.

This page can be used for rough work on any question.